

G4-EVIDENCE.md — Gate G4, scored criterion by criterion (WP4.10)

*Assembled at the close of the Step 4 build queue (wp4-01..wp4-10 plus the retrofit and the wp5-00 metric freeze), and CLOSED at the ship-gate chase. Every claim below cites a test, an evidence document, or a sealed artifact. **Verdict: ALL SIX CRITERIA MET — Gate G4 CLOSES 2026-08-02.** Criterion 1 was open at assembly (76.7%) and closed by the chase: fifteen live runs, pipeline v2 ratified pre-hoc (AM-2026-08-02-004), and the binding pair rule met at **96.7% / 100.0%** (runs 14/15, identical configuration). An earlier lucky single pass was refused; the pair rule exists because of it.*

The six criteria (plan DoD, verbatim in substance)

1. Tier-1 deterministic at calendar slots; Tier-2 $\geq 95\%$ first-pass with retry/fallback exercised. — MET (closed by the chase; assembly-time state preserved below for the record). Tier-1: deterministic by test (same tape, same words; schedule deterministic under declaration reordering). Tier-2: the retry/fallback path is exercised and logged (five live runs, zero fallbacks in the last three, every retry recorded) — but the first-pass rate is **76.7% against the frozen 95% bar** (`AUTHORING-REGRESSION.md` ; trajectory 3.3% $\rightarrow$ 76.7% across five runs, gate lineage `gate-impl/1.0.0 $\rightarrow$ 1.0.2` , prompts `letter@1.1` / `note@1.2`). **Consequence, enforced: live-world Tier-2 authoring is disabled.** Next levers, recorded in the manifest: few-shot exemplars; a pre-gate self-check pass. Iterated by version bump with full re-runs, as always.

2. No artifact in any regression run references a real firm or person, a future dateline, or a number absent from its payload's claims table. — MET. Enforced by G1/G3/G4 (blocking); all 30 run-5 recorded artifacts re-gate clean offline in CI forever (`tests/test_authoring_regression.py::TestRecordedReplay`); injection attempts that would breach it are caught (`tests/test_injection.py`).

3. Live mode runs with sealed reveal; chronicle append-only; replay reproduces the identical artifact sequence. — MET. to seal with tamper detection to one mark in one millionth (`tests/test_live_mode.py::TestSeal`); the information wall is structural (unrevealed months never enter the object); the chronicle is append-only at trigger and repository layers (Step 0, retested); replay reproduces the sequence from the chronicle, never by re-prompting (G9 records: payload hash, prompt version, model id, gate result, retry count).

4. AI committee decisions bounded, briefed only on revealed information, filed with rationales, benchmarked against heuristic ablations. — MET, with the human half deferred. Bounded (typed contract; off-menu rejects with the rejection filed), briefed behind the

structural wall, filed (no rationale, no decision; minutes to the wire), benchmarked against heuristic / random-within-bounds / hold-course on identical worlds and seeds (`ACTOR-VALIDATION.md`). The **human-cohort benchmark is deferred by owner decision D-K4-5** — a scope decision on the record, not a silent gap.

5. Documented pathology measurements exist, published internally before any client-facing claim. — MET within scope. Measured and published: fallback rate 0.20 (action-level fidelity), persona/prompt sensitivity 0.78, effect sizes structurally chained to their dispersion. The **too-rational pathology is not measurable without human cohorts** and is recorded as such — no proxy number. The no-client-facing-claim rule stands in the evidence verbatim.

6. The GenAI governance pack is complete. — MET. `governance/prompt-registry.yaml` (four prompt families, versions, hash pins, regression links, ship status); injection testing on all payload paths (authoring + committee; template engine proven single-pass); output-variance policy (Tier-1 zero by test; Tier-2 measured, never pretended away); fictional-entity enforcement (B1 screen, version recorded; G4 closed world); `governance/eu-ai-act-mapping.md` with the NIST AI RMF scaffold; and the boundary statement below.

The boundary statement

No generative output enters the certified numeric path. Structural, twice over: the engine consumes a NumericWorld that omits narrative (Step 0 guard), and `ah/core` , `ah/gen` , `ah/port` never import `ah/artifacts` — asserted by `tests/test_artifacts_service.py::TestBoundary` on every CI run since wp4-01. Generative components read the tape; they cannot write it.

The closing line

The measurement exists. Runs 14 and 15 (`letter@1.4` / `note@1.5` , `gate-impl/1.0.6` , `author-pipeline/2.0` + `self-check@1.1`) measured **96.7% and 100.0% first-pass** — two consecutive runs over the frozen bar at identical configuration, per the binding pair rule (AM-2026-08-02-004). The full fifteen-run trajectory, every instrument failure, and the refused lucky pass are in `fixtures/authoring_regression/manifest.yaml` and the archived evidence files. **Gate G4 is CLOSED.** Step 4 ends with all four project gates resolved: G0 closed, G1 evidenced with its named `mark_lag` limitation, G2 closed with its generator caveat, G4 closed clean.

Not investment advice.